ALAN TSENG

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EDUCATION

M.Sc., Electrical Engineering and Computer Science, National Taiwan University — Aug 2021 - Feb 2024
Overall GPA: 4.12/4.3
Coursework: Blockchain and Big Data, FinTech, Computer Vision

EXPERIENCE

Software Developer Intern — Jul 2022 - Nov 2022
Line Corporation (95M MAUs worldwide, social media company in Asia) — Taipei, Taiwan

- Increased API development time by 20% and solved the data inconsistency issues across teams by designing a service utilizing Swagger, auto-gen types, and Kubernetes deploy with DNS/LB.
- Led 5 developers to build an NFT merchant management system with NextJS and Spring Boot in Hackathon.
- Gave a talk on Ethereum & dApp development in the company study group.
- Developing a large scalable system using Nx and collaborating with the global team.

SELECTED PROJECTS AND COURSES

Decentralized Ticketing System - Master's Thesis, Web3 FullStack

- Reduced price fluctuations by 98%, eliminating scalper's profit (100%), by stimulating the market, writing 3 ticketing strategies and analysis using SQL.
- Developed a full-stack web3 project that includes a smart contract, website, and mobile app using Next.js, React Native, and Solidity with 95% BDD code coverage.
- Adding Fixed Price ticket purchase feature with Apache Kafka.
- Published in 2023 IEEE International Conference on Metaverse Computing, Networking and Applications.

Rate My Professors TW - Full Stack Project

- Fulfilled scraping rate of 100 KB per second using asyncio, and aiohttp, provided an API using Go and gRPC.
- Achieved 3.8K users visited in 1 year using NextJS with course scheduler, rating system, and PWA feature.
- Developing LLM+RAG, NLP, features to Build Modern GenAI Applications and an Android App.

Drum.io - Course Project

- Won top of the class with 50+ votes by developing a trainable air drum game using TensorFlow.js, Arduino, and WebSockets.

Adaptive Driving Beam System - Undergraduate Thesis

- Award a scholarship by the Ministry of Science and Technology, Taiwan, published in the 2020 IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP) and Honored for top 1% research & graduation by solving the nighttime vehicle detection and designing an adaptive driving beam using CycleGAN, YOLO, Statistics and SVM.

Early detection of human tremor symptoms and home care services based on video and deep learning

- Developed an LSTM-RNN model using Tensorflow Functional API to distinguish between patient tremor frequency and noise.
- The model achieved an average accuracy of 94.8
- The model can distinguish between different tremor frequencies with an accuracy of 94.14

Certified Developer MLOps Professional by Intel - Course

HONORS

- **Research Excellence Award** - Honored for top 1% research & graduation.
- **Intel Andy Grove Scholarship** - Selected with 2000+ applicants worldwide for academic and leadership potential.
- **College Student Research Scholarship** - Awarded by Ministry of Science and Technology, Taiwan.

EXTRA-CURRICULAR ACTIVITIES

- **Lead of Developer Student Club** - Selected with 3000+ global applicants, led a team of 73+ students, and delivered talks on machine learning, PoseNet, Python & Colab, and Flutter.
- **ACM Competitive Programing Club Leader** - Led a group of 20 students in a discussion on programming and practice using UVa Online Judge.
- **Volunteering**
  - Provided technical support at the ART FUTURE exhibition showcasing blockchain applications.
  - Assisted with Developer Student Club (DSC) and Google Developer Groups (GDG) events at SITCON(Student Information Technology Conference) and COSCUP(Conference for Open Source Coders, Users and Promoters).

SKILLS

- Artificial Intelligence(Generative AI, LLM, RAG, NLP, CV, ML/DL, Data Analysis), Web(Frontend/Backend) Development, Cloud-Native Applications, Socket, Chrome Extension, SSG/SSR, MVC, TDD/BDD, AWS, CI/CD, Web Crawling, PWA
- Python, C/C++, JavaScript/TypeScript, Golang, Java
- LangChain, Huggingface, ChatGPT, MLflow, Tensorflow, Scikit-Learn, OpenCV, OpenGL, Numpy, Pandas, Matplotlib, Swagger, FastAPI/Flask, NodeJS, Express, Nx, React(NextJS, React Native), Selenium, Playwright, Mocha, Kafka, Spark, Kubernetes, Grafana, Android, Terraform
- NoSQL(MongoDB), SQL(PostgreSQL, MySQL), Vector(Weaviate)